

Probing deception in multi-agent RAG — de-risk results

sae-deception-multiagent-rag · Vidura · 2026-06-05 · research preview (de-risk phase)

The milestone in one line: the full Phase 1/4/5 experiment now runs end-to-end and is recorded, on a CPU-tractable MiniLM proxy standing in for Gemma 2-2B + Gemma Scope. The code is built so the GPU run is a one-line backend swap. The proxy already falsified two cheap-to-falsify assumptions and redirected the method — exactly what a de-risk is for.

120

attack contexts

(60 clean / 60 poisoned)

PASS

Phase-4 gate A2.1

(poison retrieval 1.00)

1.00

random-init control

(naive labels lexically trivial)

≈0.50

probe AUROC on hardened

factual poisoning (chance)

DELIVERED THIS MILESTONE

- ✓ Runnable `src/sae_deception` package — pluggable representation backend + Gemma+SAE stub
- ✓ 120-example PoisonedRAG + Greshake corpus (naive & hardened), hashed manifests
- ✓ Phase 1 RAG pipeline + Phase 4 attack-success measurement
- ✓ Phase 5 probes: writer/retriever/combined × 3 representations, bootstrap CIs + 2 controls
- ✓ Results, 3 figures, run manifest; 5 unit tests pass; fully reproducible
- ✓ README roadmap + ASSUMPTIONS re-audit updated

METHOD (PROXY DE-RISK)

- **Representation:** MiniLM (all-MiniLM-L6-v2) embeddings as the "activation"; TF-IDF + random-init as comparison backends.
- **Probe:** L2-logistic, 5-fold OOF AUROC, 2000× bootstrap CI.
- **Controls:** random-init (A1.1, lexical triviality) and shuffled-retriever (A3.6).
- **Two corpora:** *naive* (obvious boilerplate) vs *hardened* (single-token lie / natural injection).

THREE FINDINGS THAT CHANGE THE GPU PLAN

1 · The easy task is fake.

A representation with no trained parameters scores AUROC 1.000 on naive attacks — high scores there reflect surface tokens, not deception. The random-init control is now mandatory beside every number.

2 · A frozen probe can't catch factual poisoning.

Hardened PoisonedRAG sits at **0.17 ± 0.03** (10 seeds) — reproducibly *below* chance: the probe is actively misled. Detecting lies needs generative behaviour or internal/causal signals, not a linear probe. Injection (Greshake) stays detectable (0.83) — a different problem.

3 · "Cross-agent" adds nothing — correlationally.

Combined never beat writer-only ($\Delta = +0.01$, CIs overlap); the agents share text, so a shared feature ≠ transfer. Direction 1 must rest on the Phase-6 causal ablation.

Go-forward decision

Proceed to the GPU run with three forced changes: **(a)** headline on hardened attacks + *attack-success* labels; **(b)** ship a random-init Gemma control with every AUROC; **(c)** treat cross-agent transfer as a *causal-ablation* question, keeping the correlational probe as a baseline only. Separate "injection detection" from "factual-deception detection."

REMAINING 50% (NEXT)

- Phase 2/3 — real activation capture + Gemma Scope SAE on GPU
- Phase 6 — causal ablation of candidate transfer features
- Scale corpus to $n \geq 300$ (A2.2 power)
- Phase 7 — writeup + cohort/mentor feedback

Honest caveats: MiniLM ≠ Gemma residual stream ≠ SAE features (results bound the *method*, not the phenomenon); attack-success used a worst-case reader; corpus is 60 facts × 2 templates. · Full detail: `experiments/260605_phase5_probe_derisk/results.md` · one command: `python scripts/run_experiment.py`